Report: Quantitative GWAS of barley breeding

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1 Summary

About

- I run GAPIT on the quantitative barley traits and the genotype data from a barley population
- I also implemented the core functionality of a GWAS, expecially for understanding the linear regression of the markers

Data

- This data was provided by the MLU Halle during my master programme
- Quantitative phenotype traits and a genotype matrix for multiple breed lines and multiple markers of Chromosome 2 in barely

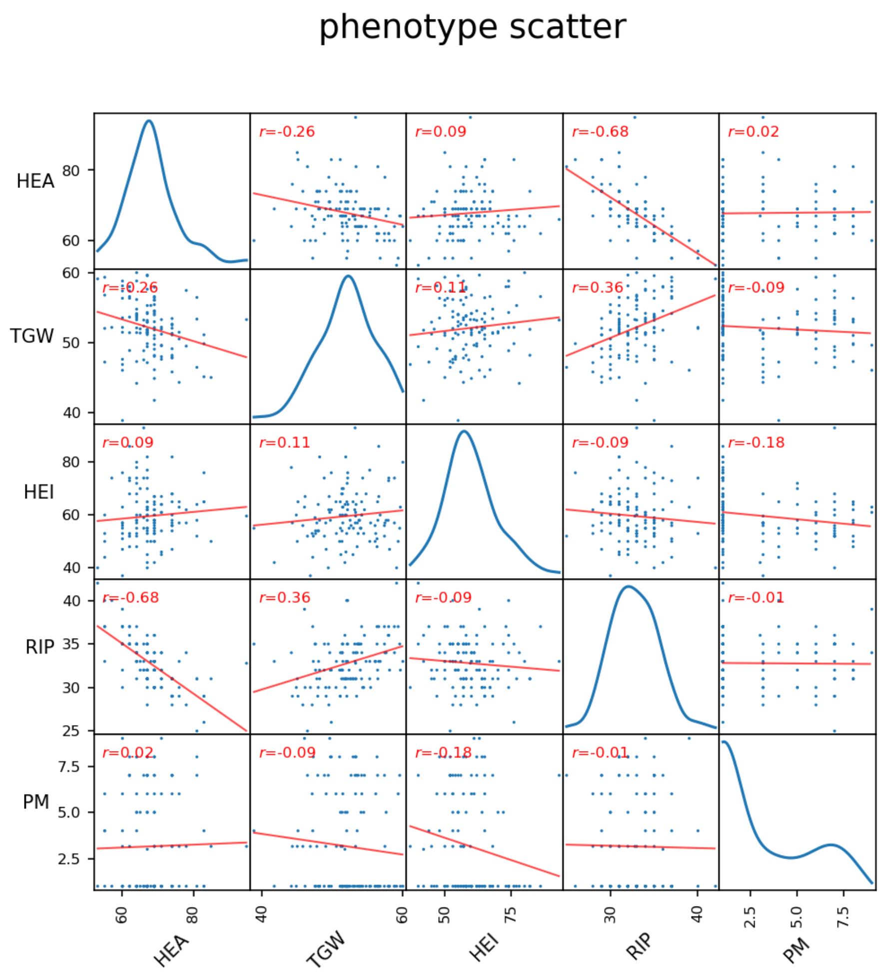
HEA	= Heading date	(Ährenschieben)
TGW	= Thousand Grain Weights	(Tausendkorngewicht)
HEI	= Height	(Höhe)
RIP	= Ripening time	(Reifedauer)
PM	= Powdery Mildew Susceptibility	(Mehltauafälligkeit)

Quantitative Traits (Phenotype)

- Normally distribution of HEA, TGW, HEI and RIP (potential poly genetic orchestration)
- Bimodal distribution of PM, hint for a genetically distinct groups in the population
- Strong anti correlation between HEA and RIP
- Weak correlation between TGW and RIP
- Weak correlation between TGW and HEA

GWAS-Manhattan

2 Phenotype



KDE

- All properties beside the PM are almost normally distributed (unimodal)
- These properties seems to undergo a complex genetic orchestra
- Instead, PM shows a bimodal shaped distriubtion, hence one might argue that there exist two distinct populations (or genetic patterns), the majority has a strong resisitance where a minority is more susceptible

TGW $\times$ RIP

- If the desired trait is a greater croop (yields more food), one can clearly see that TGW (Thousand Grain Weight) slightly correlates with the RIP (Ripening Time)
- The crop might have more time accumulating consumable nutrients (sugars, proteins, lipids)
- The scatter also shows that one would select crops that ripen later

TGW $\times$ HEA

- TGW slightly anti-correlates with HEA (Heading Date)
- But the scatter doesnt show that very clearly
- One might select for early headers

HEA $\times$ RIP

- Strong anti correlation
- High chance of a co-regulated gene orchestration

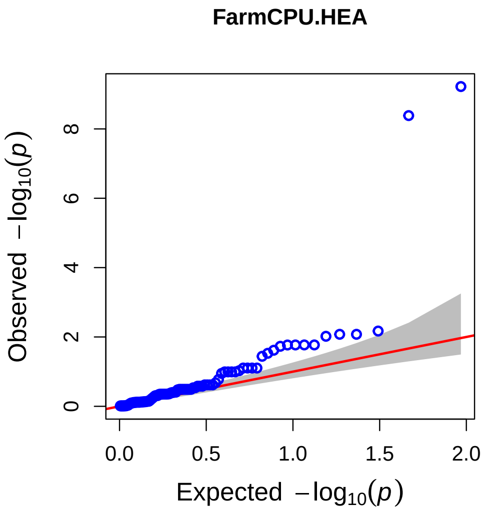
HEI $\times$ PM

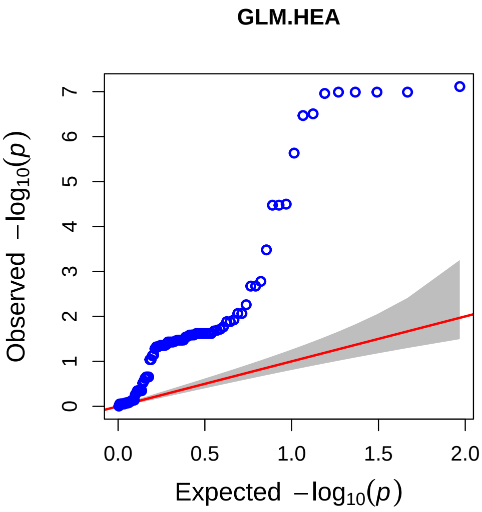
- Maybe not relevant for breeding, but interesting behavior of barley, when they have weaker resistance against powder mildew, the crops heights reduce varaince and shrink

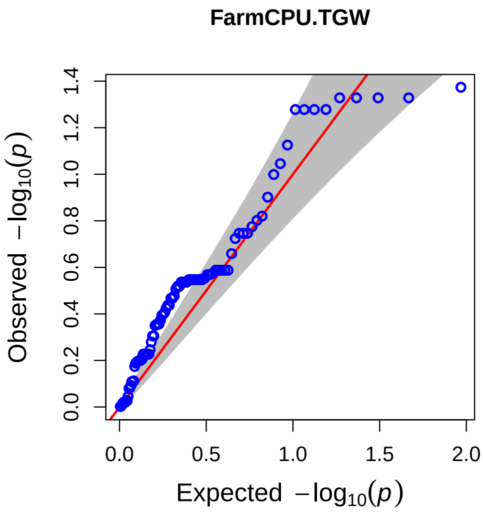
3 GWAS

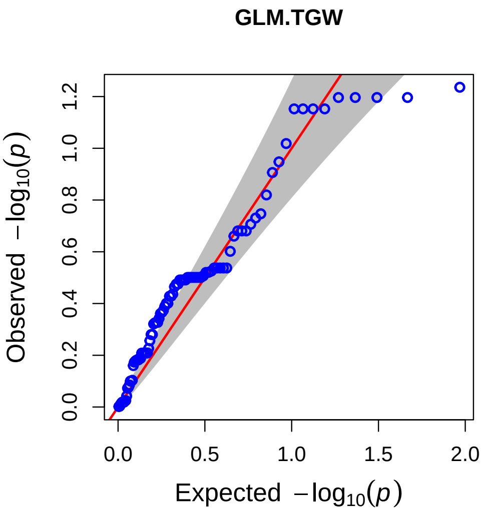
3.1 Model Evaluation

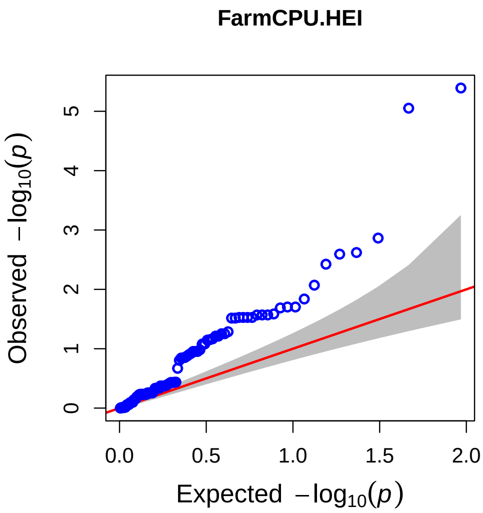
Since I used two different models, I compare the QQ plots to pick the best model for this data.

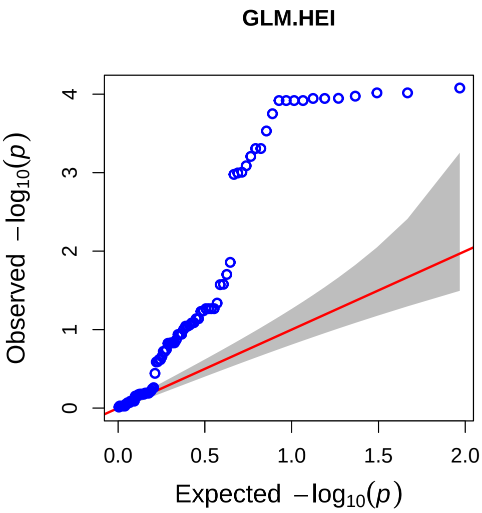


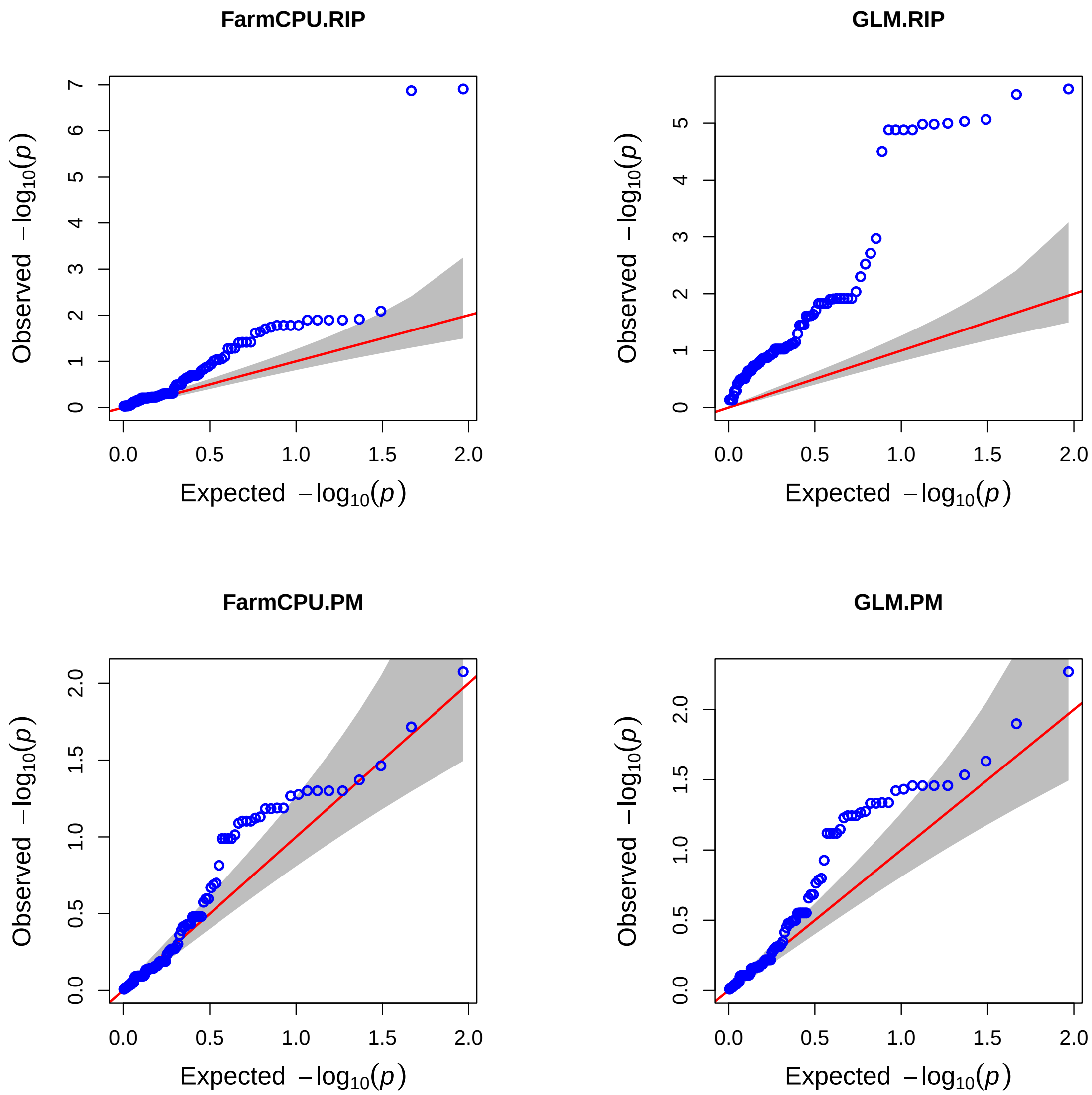










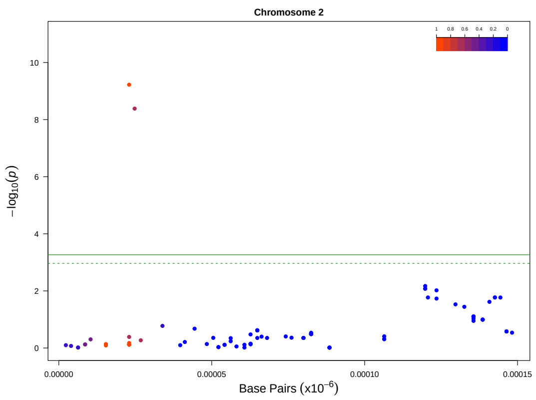


- FarmCPU outperforms the GLM w.r.t. HEA, HEI, RIP
- Both models perform equally w.r.t. TGW and PM
- The GLM shows multiple observed p values that doesnt align with the expected (false-positives), but FarmCPU shows just a few outlier, this indicates that they are supposedly true-positives
- I therefore use the FarmCPU as reference

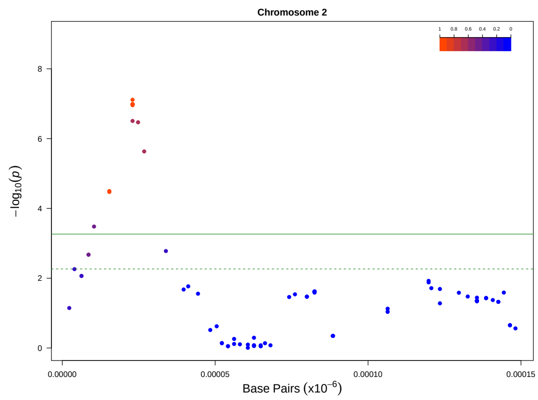
3.2 Manhattan Plots

FarmCPU (NYC, GAPIT) × GLM (NYC, GAPIT)

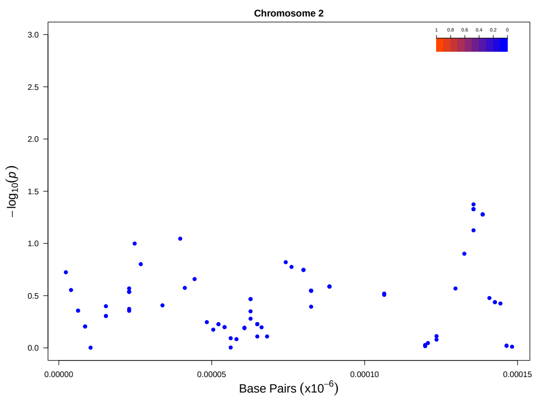
HEA - FarmCPU



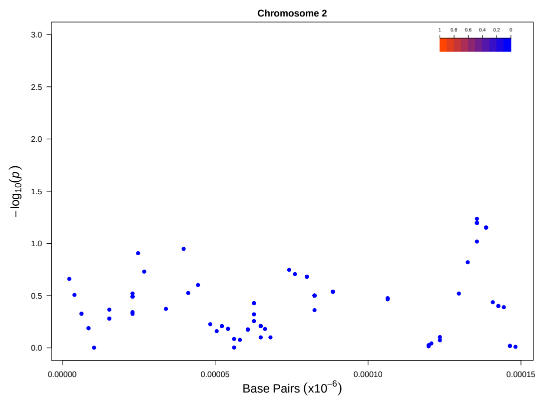
HEA - GLM



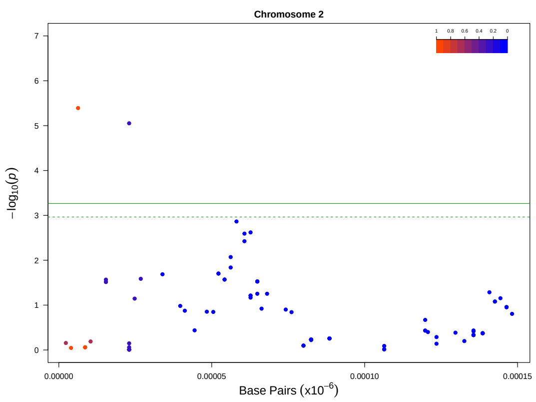
TGW - FarmCPU



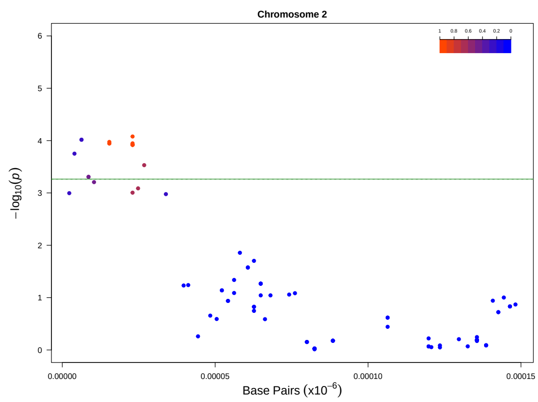
TGW - GLM



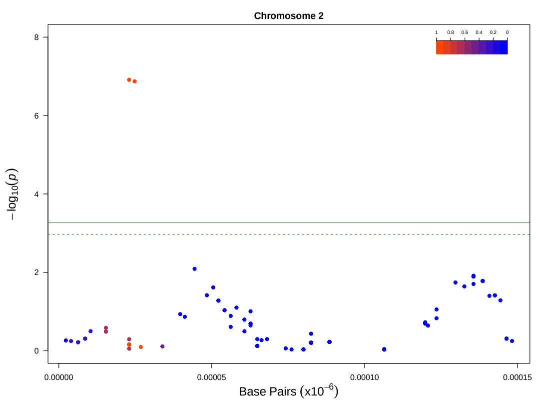
HEI FarmCPU



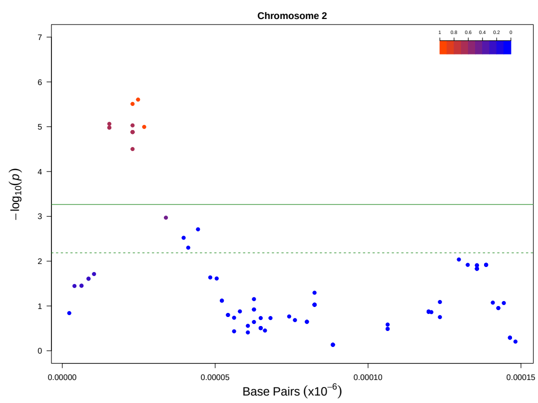
HEI - GLM



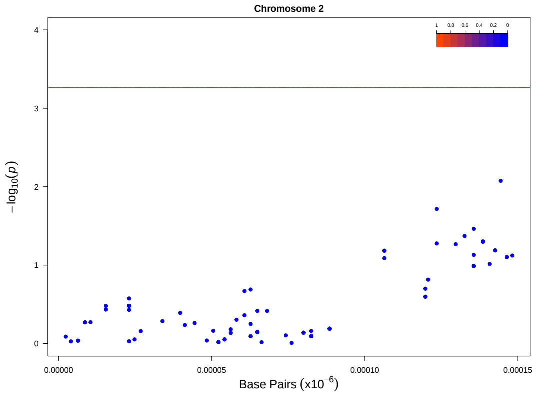
RIP FarmCPU



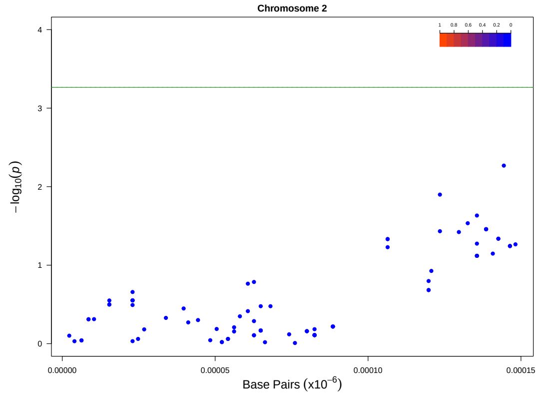
RIP - GLM



PM - FarmCPU



PM - GLM



3.3 Analysis

HEA

- Both model outputs are similiar (aligns with QQ)
- FarmCPU identifies 2 Maker to be highly significant with a high R^2 (positional close)
- Same region as GLM

TGW

- Both models outputs identical (aligns with QQ)
- TGW outputs aligns with the expected poly genetic orchestration of this quantitative trait
- Meaning that no marker is surprisingly significant

HEI

- Both model outputs are similiar (aligns with QQ)
- FarmCPU identifies 1 Maker to be highly significant with a high R^2
- FarmCPU identifies 1 Maker to be highly significant but low R^2
- Same region as GLM

RIP

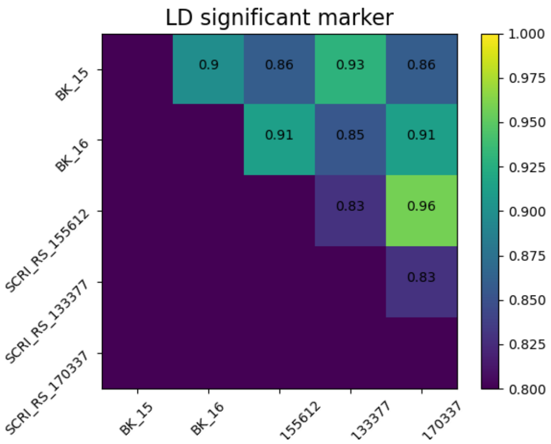
- Both model outputs are similiar (aligns with QQ)
- FarmCPU identifies 2 Maker to be highly significant with a high R^2 (positional close)
- Both Markers seem to align with the markers found in HEA trait
- These markers could correspond to the anti correlation found previously
- Same region as GLM

PM

- Both models outputs identical (aligns with QQ)
- One can clearly see that a wider genome region ($0.10 - 0.15cM$) has a genetic pattern for the powdery mildew susceptibility (with ok'ish significance $p \approx 0.05$, but not corrected)
- One could dig into this pattern, since it might align with the phenotype distribution (not relevant for this report)

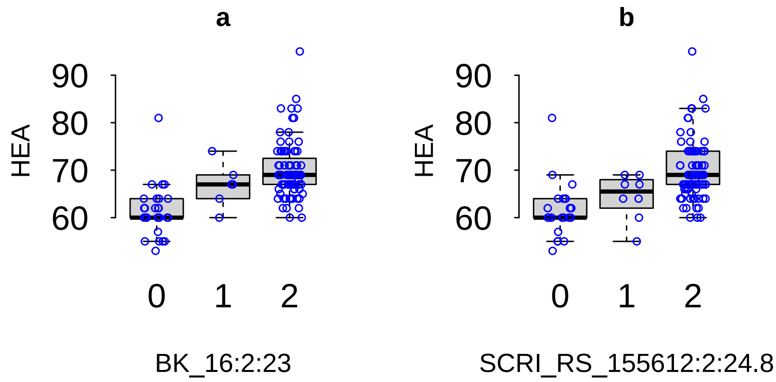
3.4 Significant Marker

Linkage Disequilibrium (LD)



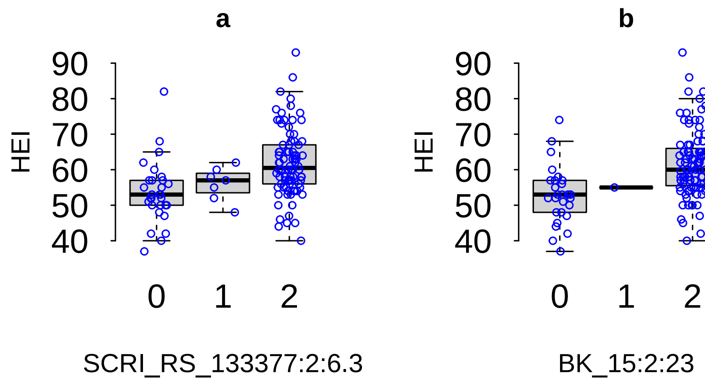
- All significant marker pairs show a $LD > 0.8$, meaning that they are non-random associated markers (co-regulate traits)
- This implies that a selection of an quantitative trait, based on one of these significant markers, also triggers another traits
- This also implies that the effect are additive

Phenotype Distribution - HEA



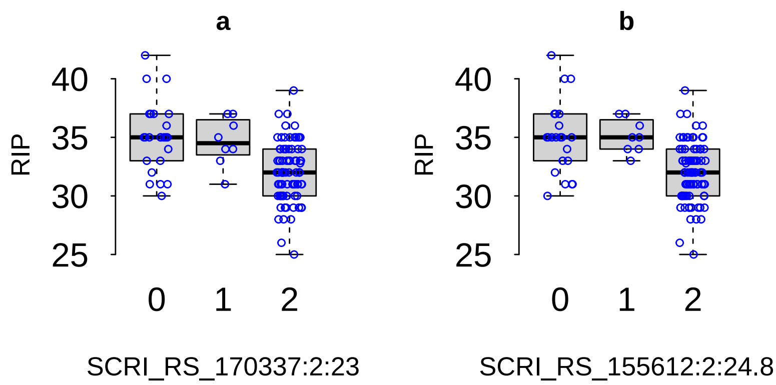
- α(BK_16) = +4.2
- α(SCRI_RS_155612) = +1.37
- Both Marker are separated by 1.8 cM

Phenotype Distribution - HEI



- α(SCRI_RS_133377) = +2,62
- α(BK_15) = +4.42
- Both Markers are separated by 16.7 cM

Phenotype Distribution - RIP



- α(SCRI_RS_170337) = -0.59
- α(SCRI_RS_155612) = -1.62
- Both Marker are separated by 1.8cM

Remark

- BK_16 (HEA) = BK_15 (HEI) = SCRI_RS_170337 (RIP) = 2.3 Mb

$\rightarrow$ Selection of these Markers effects 3 traits (pleiotropy)!

- SCRI_RS_155612 (HEA) = SCRI_RS_155612 (RIP) = 2.48 Mb (identical)

$\rightarrow$ Selection on this Marker effects 2 traits (pleiotropy)!

4 Marker-assisted selection (trait prediction)

Formula:

$$\boxed{y = \mu + \alpha_{marker} \cdot SNP + \epsilon}$$

- μ = Phenotype (A)
- α_{marker} = Effect at Marker
- SNP = Genotype at SNP (or Marker)
- Error

Potential Targets

Marker 1	SCRI_RS_155612	24.9cM	expected pleiotropic effect
Marker 2	BK_15, BK_16, SCRI_RS_170337	23.0cM	expected pleiotropic effect (Multi-Marker complex)
Marker 3	BKSCRI_RS_133377	6.3cM	single trait

Predictions by Marker

	HEA (in days)	HEI (in cm)	RIP (in days)	TGW (in g/1000 grains)	PM (in score)
Marker 1 (SNP=B)	+2.74	–	-3.24	–	–
Marker 2 (SNP=B)	+8.40	+8.84	-1.58	–	–
Marker 3 (SNP=B)	–	+5.24	–	–	–

Note: Multi-marker selection may show additive effects due to high LD between markers.

5 Breeding-Implications

Grain production

- The data reveals that Allele A seems to be in favor for TGW (production), because it implies earliest heading and latest ripening in two markers, positivly effecting TGW (compare to a B Allele)

Animal Food Production

- One might select for Marker 3 = Allele B, to get higher crops, with an acceptable effect on the other traits (from LD)